

Roll No.

TBT-501

**B. TECH. BIOTECHNOLOGY/B. TECH. BIOTECHNOLOGY
(HONS.) (WITH SPECIALIZATION IN COMPUTER SCIENCE
AND BIOLOGY)
(FIFTH SEMESTER)**

MID SEMESTER EXAMINATION, Oct., 2022

RECOMBINANT DNA TECHNOLOGY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each question carries 10 marks.

1. (a) According to you, what are top 10 discoveries in RDT to bring it into its present shape ? (CO1)

OR

- (b) What do you understand from Biosafety ? Describe layout of BSL2 facility with important features. (CO1)
2. (a) What is gene cloning ? What are different tools used in cloning ? (CO1)

OR

- (b) Discuss the design of BSL3 facility with important characteristics to prevent contamination. (CO1)

P. T. O.

(2)

3. (a) Write down the properties and types of restriction endonucleases. (CO2)

OR

- (b) What is exonuclease ? Differentiate between exonuclease and endonuclease. (CO2)

4. (a) Write short notes on the following : (CO2)

- (i) Polynucleotide kinase
- (ii) Hybrid vectors

OR

- (b) How can the recircularization of cut vector be minimized ? (CO2)

5. (a) Discuss the cloning and screening strategies in pUC18 vector. (CO2)

OR

- (b) Write short notes on the following : (CO2)

- (i) λ -vector
- (ii) YAC vector